

Geopolymer bricks made from less active waste materials

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HIGHLIGHTS

- Investigated the feasibility of producing geopolymer bricks from the less-active wastes of aggregate production industries.
- Main factors are NaOH concentration, Ca(OH)₂ content, and curing temperature.
- A Higher NaOH concentration results in higher compressive strength, less porous, and more amorphous matrix.
- Incorporation of Ca(OH)₂ increases compressive strength but decreases bulk density at larger contents.
- The compressive strength of geopolymer bricks enhances by increasing curing temperature from 70 °C to 105 °C.

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ABSTRACT

The current study aims at investigating the production feasibility of the geopolymer bricks using the waste materials emanated from the washing process of sand and gravel in aggregate industries. Geopolymer is a material originated by inorganic polycondensation as a result of the alkali activation of aluminosilicate materials. Geopolymer bricks seem to be advantageous due to the low demand for energy and the significant incorporation of wastes. The production of geopolymers from pozzolanic or aluminosilicate rich materials is very common. However, few studies have been carried out on the less-active materials such as waste materials of aggregate industries as raw materials for producing geopolymers. For this purpose, the influence of three parameters, sodium hydroxide concentration (4, 8 and 12 M), calcium hydroxide content (5%, 10%, and 15%), and curing temperature (70 °C and 105 °C), on the physical and mechanical properties of geopolymer bricks was investigated using compressive strength, SEM micrographs, XRD analysis, water absorption, and bulk density. The results indicate that higher sodium hydroxide concentration caused forming a less porous and a more amorphous microstructure, and yielded higher strengths and lower water absorption. The mixtures with higher calcium hydroxide content had higher compressive strengths (up to 75 MPa and 36 MPa at dry and wet conditions, respectively) and a more stable microstructure, however, incorporation of calcium hydroxide at levels higher than 20% results in lower compressive strengths, lower densities, and higher water absorption contents. Increasing curing temperature from 70 °C to 105 °C increased significantly the compressive strength.

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1. Introduction

Production of bricks at elevated temperatures is an energy-intensive process, which is widely used in many countries of the world. During the production of fired bricks, considerable amounts of fossil fuels are consumed, leading to CO₂ and NO_x emission in the atmosphere. On the other hand, producing bricks from clay as the raw material may cause the depletion of clay resources. Notably, clay-based bricks are one of the most widely used

materials in construction projects [1]. Thus, the production of fired bricks from clay resources is not environmentally friendly due to the high energy demand and consumption of natural resources.

The utilization of wastes as raw material for producing bricks could provide a solution with economic and environmental benefits. It helps to reduce the disposal costs in landfill sites, to maintain a sound environment [2] and also to reduce the consumption of clay resources and related costs of quarrying operations [3]. Several researchers have investigated the feasibility of brick production using a wide variety of waste materials as a replacement or an additive. They have mostly utilized rich sources

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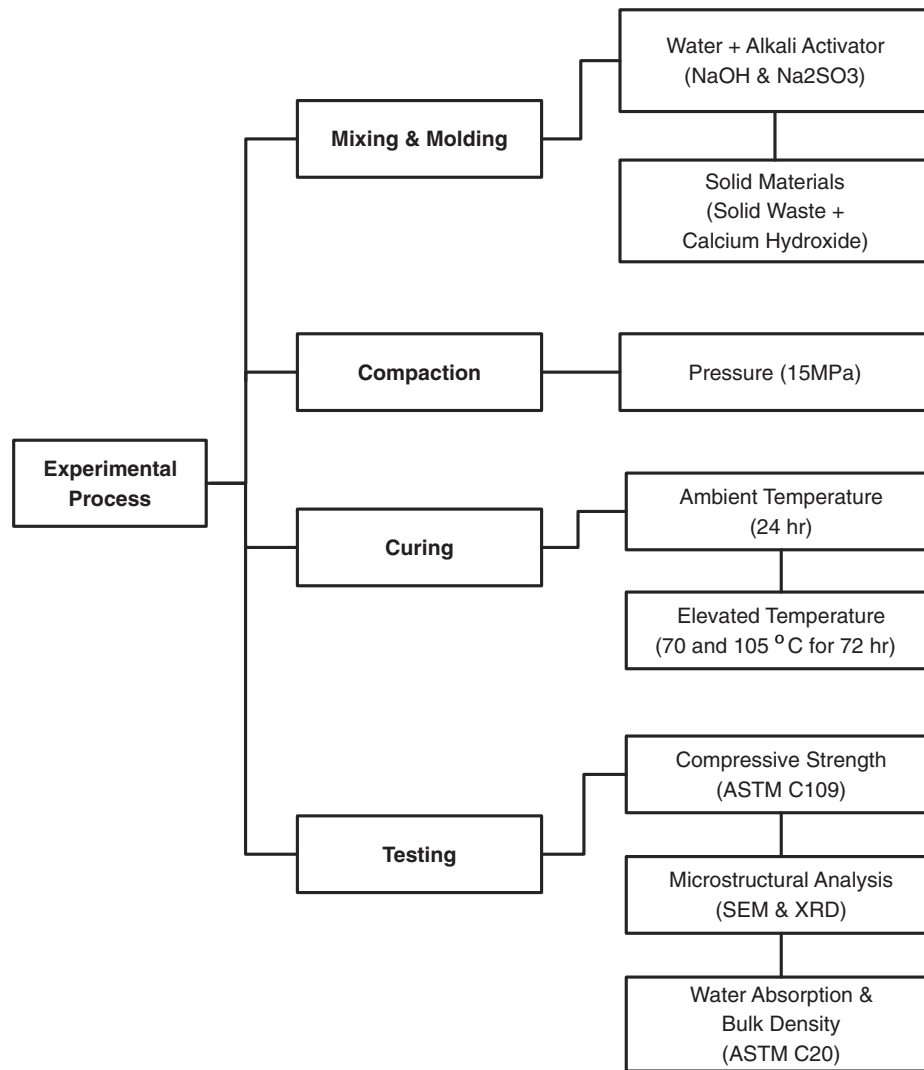


Fig. 1. Experimental process.

of aluminosilicate materials such as fly ash or slags [4–13]. Other examples of waste materials that are also used to produce bricks include mine tailings [14–16], rice husk ash [17], cotton waste [18], oyster-shell [2], and wood sawdust [19].

The waste used in this study as the raw material of geopolymer mixtures has been obtained from aggregate industries. Aggregate washing process produces large amounts of slurry during the production of sand and gravel. In order to recycle the water, the effluents should be sent into the holding ponds where the suspended solids settle down at the bottom. Lack of space for settling ponds and handling of solid waste materials are some of the problems that aggregate industries need to deal with [20].

As mentioned earlier, the high energy demand of the firing process is the main disadvantage of the fired bricks. In recent years, human has surveyed alternative solutions for producing bricks. Among them, geopolymerization may be considered as a possible technique. A geopolymer is an inorganic polymer which is produced from the polymerization of aluminosilicate raw materials using an alkali hydroxide or silicate solution [3,17,21]. Geopolymerization process includes the dissolution of silica and alumina species of the raw materials in effect of the highly concentrated alkaline solution, the formation of aluminosilicate gel, and condensation of the gel with polymerization of silica and alumina in three-dimensional polymeric structures [14,22]. Geopolymer is

Table 1
Chemical analysis of the waste soil.

Component	%
SiO ₂	53.88
Al ₂ O ₃	6.78
Fe ₂ O ₃	6.3
CaO	11.4
MgO	4.0
Na ₂ O	1.06
K ₂ O	2.34
P ₂ O ₅	0.14
TiO ₂	0.44
MnO	0.16
L.O.I	13.48

considered as an environmentally friendly material with low energy consumption, low toxicity, stability at high temperature, and high durability [23].

The motivation of the current study is to investigate the feasibility of the production of the geopolymer bricks utilizing the solid waste materials of aggregate industries. Geopolymer mixtures have been widely produced using aluminosilicate rich raw materials, however, the current study has been focused on the properties of geopolymer bricks made from less-active waste materials of

Table 2

Mixture proportions of geopolymer bricks.

Labels	Sodium hydroxide concentration (M)	Alkali solution/solid materials	Sodium silicate solution/Sodium hydroxide solution	Curing temperature (°C)	Calcium hydroxide (%)
G 4-0-70	4	0.35	2.5	70	0
G 4-10-70	4	0.35	2.5	70	10
G 4-20-70	4	0.35	2.5	70	20
G 4-30-70	4	0.35	2.5	70	30
G 8-0-70	8	0.35	2.5	70	0
G 8-10-70	8	0.35	2.5	70	10
G 8-20-70	8	0.35	2.5	70	20
G 8-30-70	8	0.35	2.5	70	30
G 12-0-70	12	0.35	2.5	70	0
G 12-10-70	12	0.35	2.5	70	10
G 12-20-70	12	0.35	2.5	70	20
G 12-30-70	12	0.35	2.5	70	30
G 4-0-105	4	0.35	2.5	70	0
G 4-10-105	4	0.35	2.5	105	10
G 4-20-105	4	0.35	2.5	105	20
G 4-30-105	4	0.35	2.5	105	30
G 8-0-105	8	0.35	2.5	105	0
G 8-10-105	8	0.35	2.5	105	10
G 8-20-105	8	0.35	2.5	105	20
G 8-30-105	8	0.35	2.5	105	30
G 12-0-105	12	0.35	2.5	105	0
G 12-10-105	12	0.35	2.5	105	10
G 12-20-105	12	0.35	2.5	105	20
G 12-30-105	12	0.35	2.5	105	30

aggregate production plants. It is worth mentioning that waste materials of aggregate industries are accessible in many parts of the world. The results could be helpful for future developments in the industrial production of geopolymer bricks.

As well as raw material characteristics, many factors, such as molarity of sodium hydroxide, water glass (sodium silicate) to sodium hydroxide ratio, liquid to binder ratio, forming pressure and curing conditions influence physical and mechanical properties of geopolymer bricks such as dimensionality, density, water absorption, initial rate of absorption (IRA), mechanical strength (compressive, flexural, shear and bond) and microstructural properties [24]. In this study, compressive strength, water absorption, bulk density, and microstructural analysis including SEM imaging and XRD analysis have been conducted to investigate the influence of three factors including sodium hydroxide concentration, calcium hydroxide content, and curing temperature on the characteristics of geopolymer bricks.

2. Experimental study

The flowchart for the experimental process is depicted in Fig. 1.

2.1. Materials

The raw material used in this research was sourced from waste materials of aggregate industries. The solid waste material was passed through No. 50 (300 μ m) sieve before mixing. The chemical analysis of solid waste is presented in Table 1. Other materials include alkali-activator, which is sodium hydroxide and Water Glass (sodium silicate) with 3.1 M ratio and 41% solid content, hydrated lime (calcium hydroxide), and tap water.

2.2. Mixture proportions and specimens preparation

The mixing procedure used in this study consists of: i) mixing the waste material and calcium hydroxide with a mechanical mortar mixer for 2 min, ii) adding sodium hydroxide solution and sodium silicate solution to the dry ingredients, iii) mixing for 2 min followed by a 1 min stopping and, iv) mixing the paste for another 2 min. Afterward, the specimens were compressed to form

cubic dimensions of 50x50x50 mm using a hydraulic compression machine. The forming pressure was about 15 MPa for each specimen. The specimens were then covered to minimize water evaporation. After 24 h being at ambient temperature, the molds were stripped, and the specimens were cured in an oven at 70 °C or 105 °C under a plastic cover for a period of 48 h. Finally, the specimens were uncovered and dried in the oven for another 24 h.

In this research, the influence of three factors, sodium hydroxide concentration, calcium hydroxide content, and curing temperature was studied. Other parameters, including sodium silicate to sodium hydroxide ratio, and liquid to solid materials ratio, were kept constant at levels of 2.5 and 0.35, respectively. Several studies have suggested the value of 2.5 for the sodium silicate to sodium hydroxide ratio in alkali-activated geopolymer mixes [10–13,25]. It has also been observed that the ratio of liquid to solid materials in clay-fly ash geopolymer mixes with a content of lower than 0.3 or higher than 0.8 is not appropriate [9]. The suggested ratios in several researches were 0.42–0.58 [26], 0.4 [27], 0.3 [11,28], 0.35 [5], 0.08–0.18 [14], 0.12–0.2 [16], 0.5 [10], 0.2 [12], 0.9 [13]. In this study, the values of 2.5 and 0.35 were selected for these two ratios based on the initial experiments to produce mixes with sufficient workability to form the desired shape without initial minor cracks in the specimens.

Mixture proportions, in this study, are presented in Table 2. As can be seen, the specimens were prepared at 4, 8, and 12 M sodium hydroxide concentrations, 70 and 105 °C curing temperatures, 0 to 30 percent calcium hydroxide contents, and were cured under dry and wet conditions. In the wet condition, specimens were placed in water for 24 h after the elevated curing condition in the oven. It should also be noted that an appropriate value for the sodium hydroxide concentration, based on the findings in literature, is in the range of 5 to 15 Molar [5,10,31,11–14,16,28–30]. Regarding the curing temperatures, alkali-activated mixes can be made either at ambient or high temperature. Several researchers reported curing temperatures in the range of 23 to 120 °C [5,9,10,12,14,27–30,32]. However, it is recommended to use 100 °C as an upper limit [24]. According to these studies, elevated temperatures can substantially increase the development in the properties of geopolymers. In the current study, several initial mixes were produced to examine the influence of curing temperature and its duration on the compressive

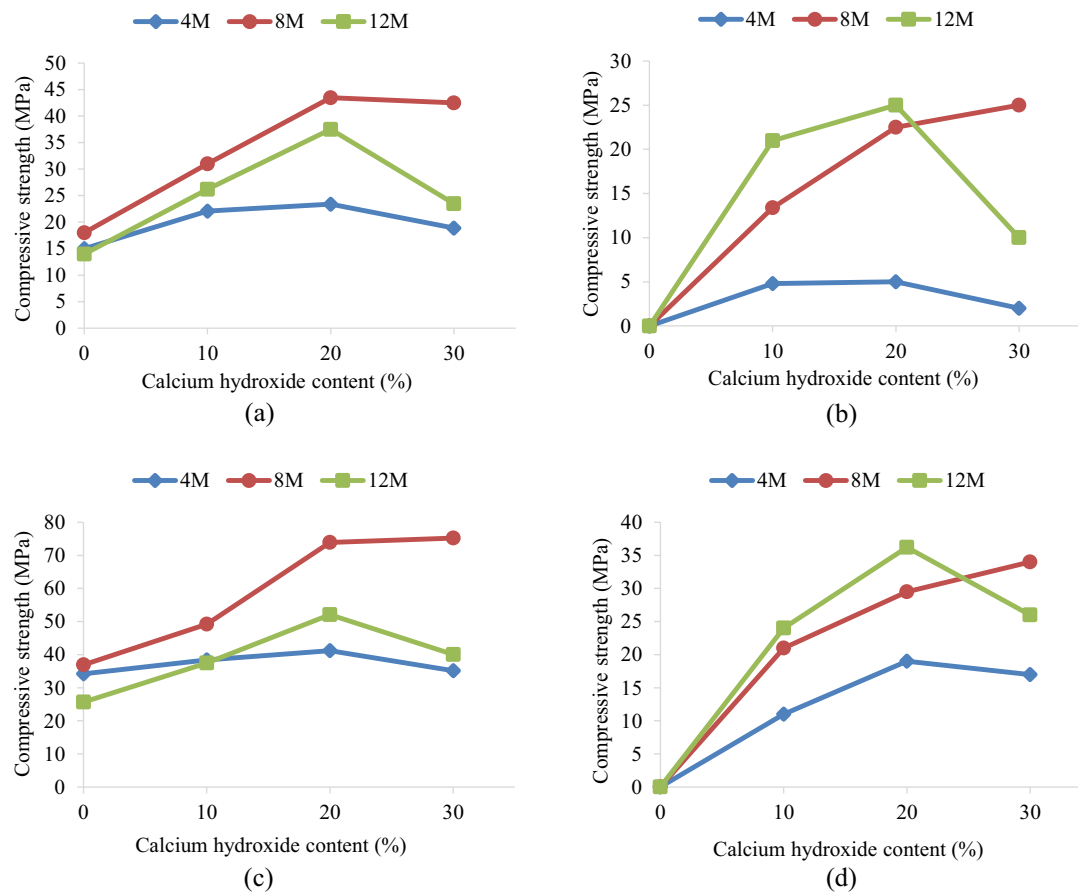


Fig. 2. Effect of sodium hydroxide concentration and calcium hydroxide content on the compressive strength for: (a) dry and cured at 70 °C (b), wet and cured at 70 °C, (c) dry and cured at 105 °C, and (d) wet and cured at 105 °C.

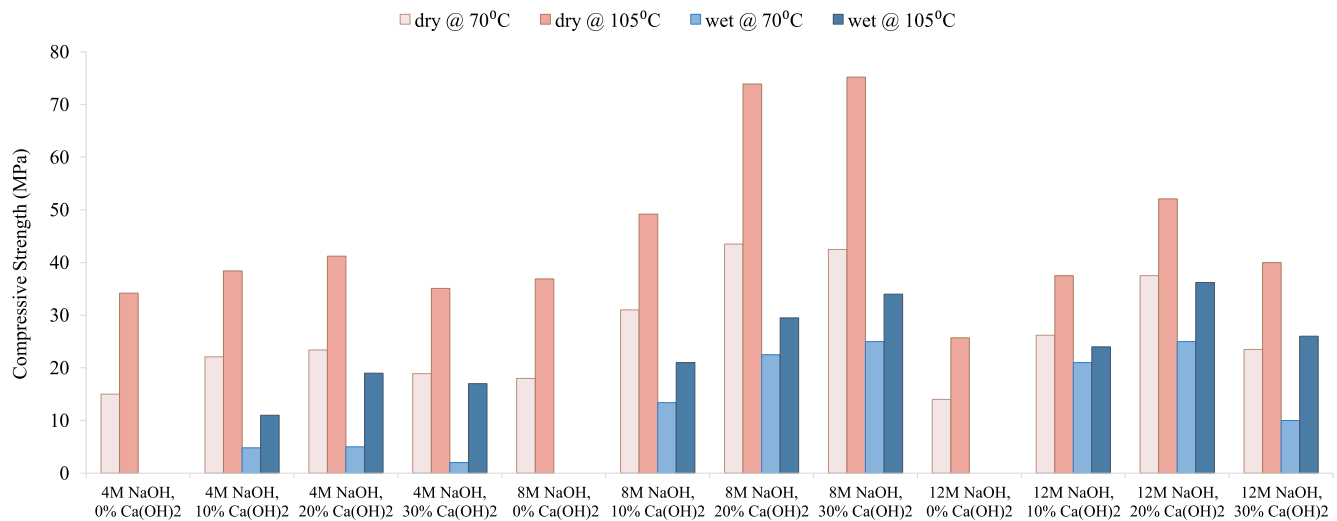


Fig. 3. The effect of temperature and curing conditions on compressive strength.

strength of the specimens. Based on the obtained results, it has been deduced that the utilized 70 °C, and 105 °C temperatures are more appropriate, which could reflect the differences between the curing conditions. The temperatures lower than 70 °C were not suitable because these temperatures could not improve the mechanical properties significantly. Higher temperature degrees than 105 °C haven't also been chosen due to the high demand for energy and the range of curing temperatures found in the previous researches.

2.3. Test methods

The tests performed in this study consist of compressive strength, SEM imaging, XRD analysis, water absorption, and bulk density determination.

2.3.1. Compressive strength

Compressive strength values were determined in accordance with ASTM C109 [33]. A hydraulic compression machine was

utilized for measuring the compressive strength at a constant loading rate of 1.8 kN/s. For each of dry and wet condition, three specimens were tested after exposing to the curing conditions, and the average compressive strength was determined.

2.3.2. Microstructural analysis

Microstructure and morphology of several specimens were investigated using scanning electron microscopy (SEM) images and X-ray diffraction (XRD) analysis. The SEM imaging was performed using the FE-SEM Hitachi S-4160 microscope, and the XRD analysis was conducted with EQUinox 3000 diffractometer.

2.3.3. Water absorption and bulk density

The water absorption was calculated as follows:

$$\text{Water absorption (\%)} = (W_2 - W_1)/W_1 \quad (1)$$

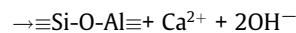
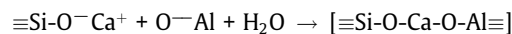
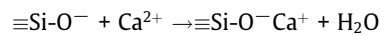
where W_1 is the weight of specimen after complete drying at 105 °C, and W_2 is the weight of specimen after 24 h of soaking.

The bulk density was also measured using the Archimedes immersion technique. Both water absorption and bulk density were conducted according to the standard method of ASTM C20 [34]. The results of water absorption and bulk density were the average results of 3 specimens.

3. Results and discussion

3.1. Compressive strength test results

Fig. 2 shows the effect of calcium hydroxide content and sodium hydroxide concentration on the compressive strength test results in the wet and dry curing conditions. All compressive strength values are the average of three results, which their coefficients of variations were lower than 12%. The results indicate that the calcium hydroxide content and sodium hydroxide concentration could significantly influence the compressive strength of geopolymer bricks. Although the obtained strength of the dry specimens at 0% calcium hydroxide content is partly a result of the drying process of sodium silicate in the mixtures, increasing the calcium hydroxide content raised compressive strengths, especially in wet condition. Higher compressive strengths of the mixtures with calcium hydroxide are mainly attributed to the formation of a more stable microstructure, which is possibly due to several reasons. First, calcium hydroxide may be bonded in a geopolymeric gel to obtain charge balance by replacing cations within the geopolymer [27]. Integration of calcium cations in the geopolymeric gel occurs via the following scheme [16]:



Second, calcium hydroxide could be converted to calcium carbonate in the mixture, which could strengthen the structure and reduce the porosity of the specimens [26]. Third, calcium hydroxide can react with sodium silicate solution and lead to formation of amorphous calcium silicate hydrate (CSH) or calcium aluminosilicate hydrate (CASH), causing a water deficiency and raising mixture's alkalinity; thus, the more alkalinity will result in higher dissolution of aluminosilicate species and consequently higher level of polycondensation and geopolymerization [32]. Moreover, the low solubility of calcium carbonate in the water contributes to the durability of the geopolymer brick specimens [16].

Nonetheless, at the incorporation level of 30%, calcium hydroxide could not enhance mechanical characteristics compared to the

specimens with 20% calcium hydroxide, and lower strengths were obtained. Short setting times and lower workability of the mixtures with higher calcium hydroxide content can lead to insufficient compaction of the specimens, and consequently causing lower strengths. This effect was evidenced during sample preparation, especially in the mixtures containing 12 Molar sodium hydroxide solution. Therefore, it seems that the incorporation of 8 Molar sodium hydroxide and 20% calcium hydroxide are the optimum concentrations to provide ideal alkalinity and form a more stable binder. On the other hand, increasing calcium hydroxide content to the values higher than 20% may affect the optimal structure of the geopolymeric gel.

Fig. 2 reveals that specimens with 8 Molar sodium hydroxide solution outperformed the other specimens at similar calcium hydroxide incorporation levels in the dry condition; however, in wet condition, the specimens containing 12 Molar sodium hydroxide solution gained higher strengths. Generally, increasing sodium hydroxide concentration may increase the solubility of aluminosilicate species, promoting geopolymeric gel formation, and consequently yielding higher compressive strength. However, a high level of sodium hydroxide will increase alkalinity, causing short setting time and insufficient compaction of specimens. It may also result in undesirable morphology and non-uniformity of the products [35]. The possible reason why specimens containing 12 M sodium hydroxide gained higher strengths in the wet condition is that higher sodium proportion increases the formation of the geopolymer gel [36]. Thus, the more formed geopolymer gel within the binder will enhance the stability and durability of the mixtures in the water. In other specimens with lower sodium hydroxide concentration, degeopolymerization of the geopolymeric gel takes place faster, and accordingly, compressive strength decreases.

The influence of temperature and curing conditions on the compressive strength is shown in Fig. 3. Increasing the curing temperature from 70 °C to 105 °C led to higher compressive strengths.

Table 3

ASTM specifications for different applications of bricks.

Specification	ASTM designation	Minimum compressive strength (MPa)	Maximum water absorption (%)
Building brick	C62-10	20.7	17
Solid masonry unit	C126-99	20.7	NA
Facing brick	C216-07a	20.7	17

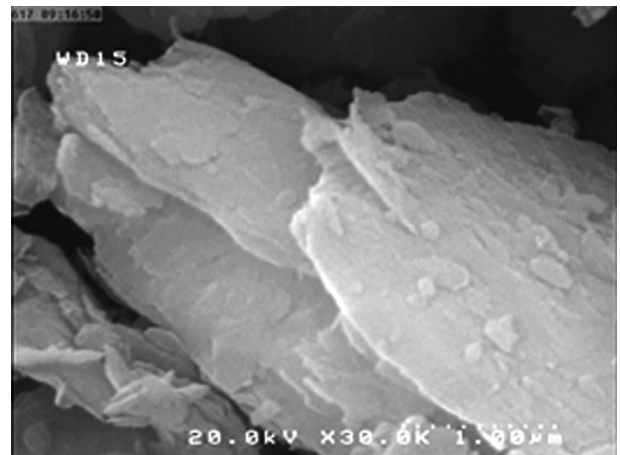


Fig. 4. SEM micrograph of waste soil powder.

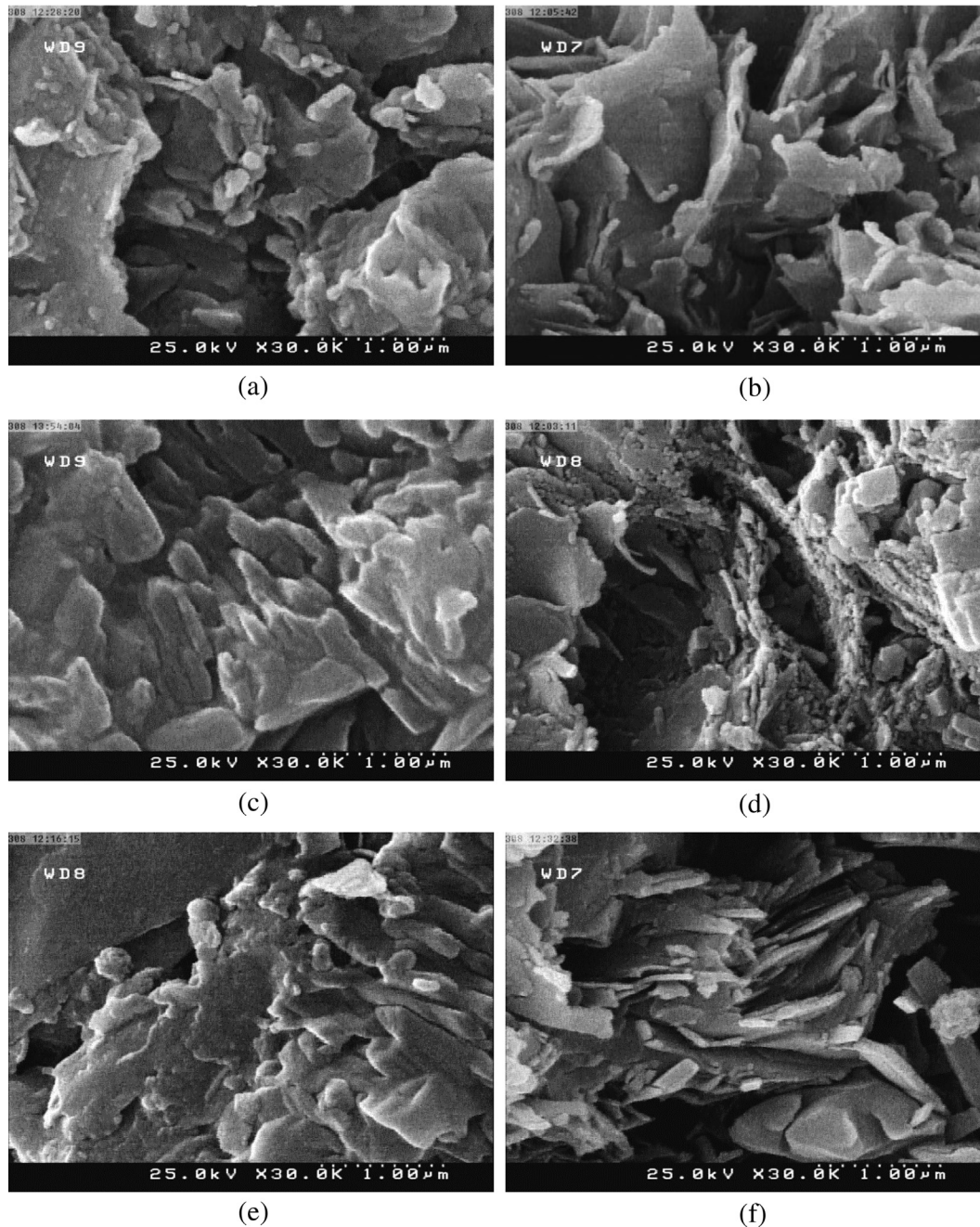


Fig. 5. SEM micrographs of (a) 4 M sodium hydroxide concentration with 0% calcium hydroxide; (b) 4 M sodium hydroxide concentration with 20% calcium hydroxide; (c) 8 M sodium hydroxide concentration with 0% calcium hydroxide; (d) 8 M sodium hydroxide concentration with 20% calcium hydroxide; (e) 12 M sodium hydroxide concentration with 0% calcium hydroxide; (f) 12 M sodium hydroxide concentration with 20% calcium hydroxide.

Higher curing temperature helps accelerate the dissolution of aluminosilicate species and then polycondensation and geopolymerization; however, without any coverage, the more elevated temperature may cause water evaporation from the geopolymer matrix and consequently further dissolution of aluminosilicates may be hindered [14]. In this study, in order to reduce water evaporation and increase the further formation of geopolymeric gel, a plastic cover has been used. The highest compressive strengths of the mixtures cured at 105 °C were obtained about 75 MPa and 36 MPa for the dry and wet curing conditions, respectively. According to the ASTM specifications, the required minimum compressive strengths of the bricks are presented in Table 3.

3.2. SEM images

SEM analysis was performed to investigate the microstructure and phase composition of the geopolymer brick specimens. Fig. 4 shows the SEM micrograph of the original waste soil powder. As it is shown, the waste material is composed of coarse crystalline particles with irregular shapes and open microstructure which do not adhere to each other. As mentioned before, this compound is produced through the washing process of aggregates, and it is very inactive in reaction with water. Thus, this material should be synthetically activated. That is why the geopolymerization technique has been employed in this study.

Fig. 5 shows the effect of sodium hydroxide concentration and calcium hydroxide content on the microstructure of the geopolymer brick specimens. The microstructure of the specimens containing 4 M, 8 M, and 12 M concentration of sodium hydroxide with 0% of calcium hydroxide is presented in Fig. 5a, c, and e, respectively. As can be seen, by increasing the sodium hydroxide concentration, a well-formed and confined microstructure is developed in which the coarse particles have been converted to finer ones and adhered to each other, and consequently an amorphous microstructure is formed, which could be due to a higher rate of dissolution of aluminosilicate species and geopolymerization reactions at higher alkalinity of the mixture. The conversion of the initial raw material to the geopolymer is evident in the SEM images.

The influence of the addition of 20% calcium hydroxide on the microstructure of geopolymer bricks is presented in Fig. 5b, d, and f. It shows that the incorporation of calcium hydroxide resulted in a more fine, and homogeneous microstructure. It appears that these compounds are more stable compared to those without calcium hydroxide. The compressive strength test results also revealed that the incorporation of calcium hydroxide contributes to the stability of the specimens.

Fig. 6a shows the microstructure of the specimen with 4 M sodium hydroxide and no added calcium hydroxide in wet condition. As can be seen, wet condition adversely affected the microstructure by dissolving the geopolymeric gel and forming wide gaps between the particles. However, as it is evidenced in compressive strength test results, incorporation of calcium hydroxide and a higher concentration of sodium hydroxide could enhance the properties of the specimens in wet condition. Fig. 6b shows how the addition of 8 M sodium hydroxide and 20% calcium hydroxide reduced degeopolymerization owing to the formation of calcium-based products, which helped to fill the pores and bridge the gaps between the different phases and unreacted particles.

3.3. XRD analysis

XRD analysis was conducted on the waste soil powder and one of the geopolymer brick specimens with 8Molar sodium hydroxide and 20% calcium hydroxide at 105 °C, as presented in Fig. 7. According to Fig. 7a, XRD analysis of waste soil powder shows that it is mainly made of quartz (Q), calcite (Ca), albeite (Al), anorthite (An), clinocllore (Cl), and illite (I). The Higher crystalline peak corresponding to quartz in the XRD pattern supports the data presented in Table 1, indicating the presence of a large amount of crystalline silica in the waste material.

XRD analysis of geopolymer brick in Fig. 7b shows that in addition to quartz, calcite, anorthite, and illite, there are other compositions such as sodium aluminosilicate (S) and diopside (D). Sodium aluminosilicate is one of the main strength-giving compounds that is formed after geopolymerization. Besides, the presence of calcite or calcium carbonate in the XRD pattern of geopolymer brick specimen indicates that calcium hydroxide is reacted with carbon dioxide in the air, and consequently, calcium carbonate is formed, which is more durable in the vicinity of the water. As mentioned before, incorporation of calcium may lead to the formation of CSH gel. However, due to the amorphous structure of CSH, no distinct peak of this compound could have been observed in the XRD patterns.

In order to have a better comparison between the crystalline peaks of XRD patterns of the waste soil powder and the geopolymer brick, the superposition of Fig. 7a and b is presented in Fig. 7. The crystalline phase of the waste soil powder is significantly transformed into an amorphous phase through the dissolution of crystalline particles in the alkaline solution during the geopolymerization process. Likewise, SEM images showed that

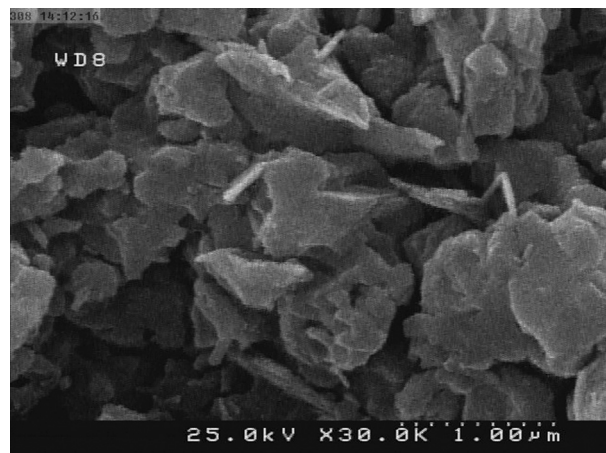


Fig. 6. SEM micrograph of geopolymer brick specimens under wet condition: (a) 4 M sodium hydroxide concentration with 0% calcium hydroxide; (b) 8 M sodium hydroxide concentration with 20% calcium hydroxide.

the geopolymerization would lead to a more amorphous microstructure. Higher compressive strengths of the mixtures confirm the microstructural analyses on creating geopolymer phases in the modified mixtures.

3.4. Water absorption and bulk density

Water absorption and bulk density test results are presented in Fig. 8a and b, respectively. Fig. 8a shows that increasing sodium hydroxide concentration led to lower water absorption, especially for the mixtures with calcium hydroxide content more than 10%. A higher degree of geopolymerization in the mixtures with a higher concentration of sodium hydroxide resulted in a less porous matrix, and consequently, lower water absorption was obtained. Apart from that, increasing calcium hydroxide content caused higher water absorption. As it is clear, water absorption values vary from 10 to 16 percent, which are below the limits of ASTM C62-10 [37] requirements, as shown in Table 3. According to Fig. 8a, larger contents of calcium hydroxide, also decreased the bulk density of the brick specimens. Gaining lower densities and higher water absorption, by adding more calcium hydroxide, especially contents more than 20%, is in agreement with abovementioned results of compressive strength that showed 30% of calcium hydroxide would decrease the compressive strength.

As presented in Fig. 8b, by increasing calcium hydroxide content, the bulk density of the specimens was reduced. The lower bulk density of the bricks containing calcium hydroxide could be due to the lower bulk density of calcium hydroxide compared to the waste soil.

4. Conclusion

Feasibility of using the waste material of aggregate industries to produce geopolymer bricks was studied by investigating the effects of three major factors, including sodium hydroxide concentration, calcium hydroxide content, and curing temperature, on compressive strength, microstructure, water absorption, and bulk density. Based on the results, the following conclusions were drawn:

The geopolymer brick specimens prepared at 8 M sodium hydroxide concentration gained higher dry compressive strengths compared to those produced at 4 M and 12 M. The maximum compressive strength at dry condition was obtained more than 75 MPa for the mixtures with 8 M sodium hydroxide concentration and

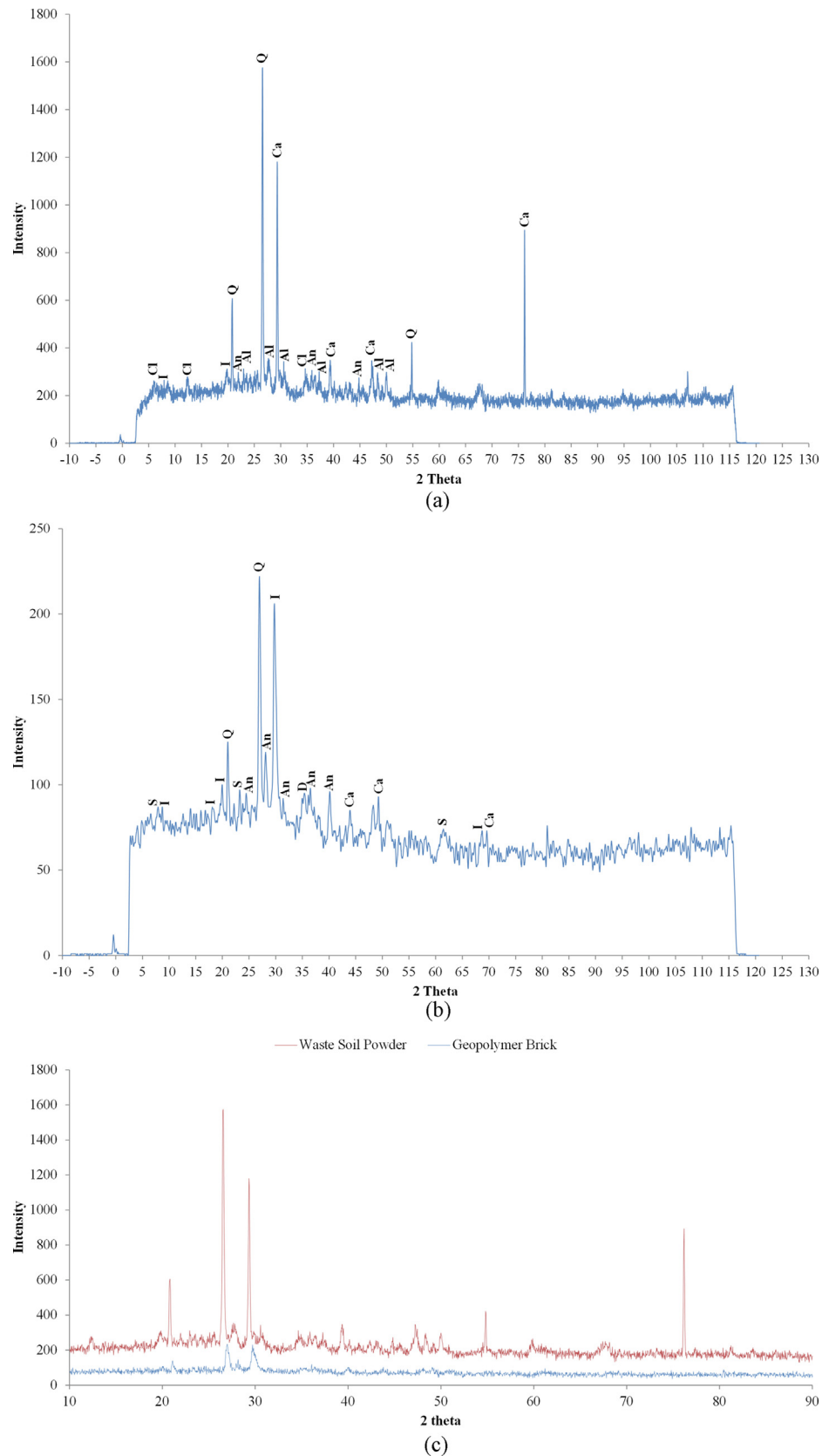


Fig. 7. XRD patterns (a) waste soil powder; (b) geopolymer brick; (c) Superposition of (a) and (b).

20% calcium hydroxide. However, in wet condition, the highest compressive strength was about 36 MPa at 12 M sodium hydroxide concentration.

Incorporation of larger contents of calcium hydroxide seems to form a more stable brick by causing higher compressive strengths, and the formation of finer microstructure; however, lower com-

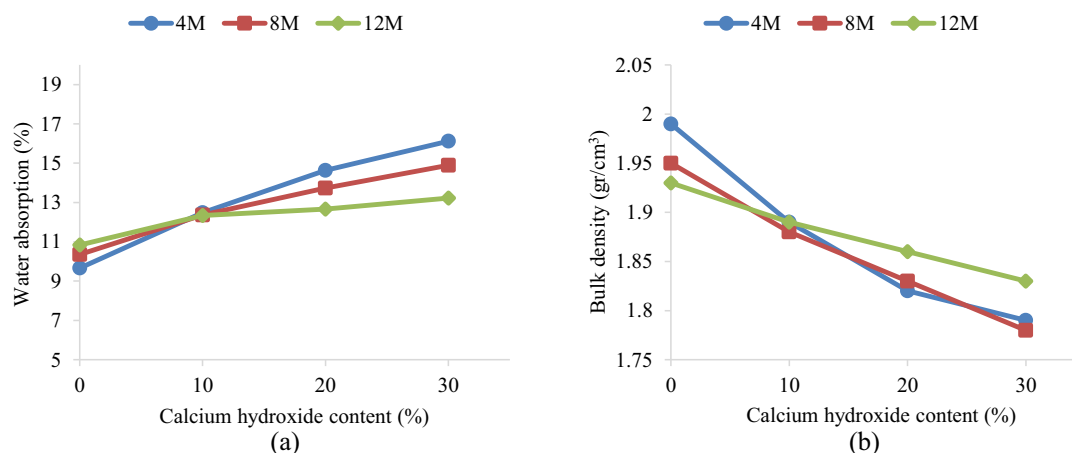


Fig. 8. Effect of calcium hydroxide and sodium hydroxide on (a) water absorption, and (b) bulk density.

pressive strength, higher water absorptions and lower densities have been resulted from the addition of calcium hydroxide content more than 20%.

Increasing the curing temperature from 70 °C to 105 °C, enhanced significantly the compressive strength of the geopolymer brick specimens by about an average enhancement of 76%.

Producing geopolymer bricks using wastes of aggregate industries is feasible, utilizing appropriate preparation conditions (sodium hydroxide concentration, calcium hydroxide content, and curing temperature), suitable liquid to solid materials ratio and sodium silicate to sodium hydroxide ratio.

CRedit authorship contribution statement

H. Madani: Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Validation. **A.A. Ramezani-pour:** Supervision, Conceptualization, Methodology. **M. Shah-bazinia:** Investigation, Writing - original draft, Visualization. **E. Ahmadi:** Investigation, Writing - original draft, Writing - review & editing, Visualization, Data curation.

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